

Roadmap for your “AI-Powered Loan Eligibility Advisor”

Phase 1: Planning & Setup

- **Define Scope & Features**
 - Finalize input features (credit score, income, loan history, employment, debt-to-income ratio, etc.).
 - Decide on output format (approval probability, risk category, eligibility decision).
 - Select ML algorithms (Logistic Regression, Random Forest, XGBoost, etc.).
- **Environment Setup**
 - Set up GitHub repo for version control.
 - Create Python virtual environment.
 - Install required libraries (Streamlit, Scikit-learn, Pandas, SHAP, ReportLab/Fpdf, Hugging Face Transformers, etc.).

Milestone: Project skeleton ready with environment & repo initialized.

Phase 2: Data Ingestion & Preprocessing

- **Data Sources**
 - Use open datasets like *loan prediction datasets* (Kaggle, UCI).
 - Optionally enrich with synthetic data for balance.
- **Data Cleaning**
 - Handle missing values, duplicates, and inconsistencies.
 - Encode categorical variables (One-Hot/Label Encoding).
 - Scale numerical features (StandardScaler/MinMaxScaler).
- **Feature Engineering**
 - Create ratios (Debt-to-Income, Loan-to-Value).
 - Categorize applicants into bins (low, medium, high credit score).

Milestone: Cleaned, well-structured dataset with train/test splits.

Phase 3: Machine Learning Model Training & Inference

- **Model Training**
 - Train baseline model (Logistic Regression) for explainability.
 - Experiment with advanced models (Random Forest, XGBoost, LightGBM).
 - Perform hyperparameter tuning (GridSearchCV / Optuna).
- **Model Evaluation**
 - Use accuracy, F1-score, AUC-ROC.
 - Validate on test set and check bias across groups.
- **Model Export**
 - Save trained model using joblib or pickle for deployment.

Milestone: Best-performing model selected & saved.

Phase 4: Explainability and Report Generation

- **Explainable AI**
 - Integrate SHAP for feature attribution.
 - Generate visual plots (feature importance, SHAP summary).
- **PDF Report**
 - Create template summarizing:
 - Applicant inputs
 - Approval probability & decision
 - SHAP visualization
 - Recommendations
 - Generate PDF with reportlab or fpdf.

Milestone: Fully automated PDF decision summary with visuals.

Phase 5: Chatbot Integration

- **Chatbot Core**
 - Use Hugging Face transformer models (DistilGPT-2, Flan-T5, etc.).

- Fine-tune prompts for financial advice (loan terms, EMI, credit score).
- **Functionality**
 - Handle FAQs like:
 - “What is EMI?”
 - “Why was my loan not approved?”
 - “How can I improve my credit score?”
 - Contextual responses based on user’s application.
- **Integration**
 - Add chatbot UI in Streamlit with conversation memory.

Milestone: Interactive chatbot assisting users in financial literacy.

Phase 6: Web Application & Deployment

- **Streamlit App Development**
 - User form for loan application.
 - Backend inference pipeline calling ML model.
 - Visualization of predictions and chatbot interaction.
- **Deployment**
 - Host on Streamlit Cloud / AWS / GCP.
 - Enable secure user authentication (optional).
- **End-to-End Testing**
 - Test with multiple scenarios.
 - Validate speed, accuracy, and usability.

Milestone: Fully functional AI-powered loan advisor deployed online.